SASHTHAM INDRIYA

AI-POWERED BREATH-BASED NARCOTICS DETECTION DEVICE

Technical Architecture • Sensing Methodology • Prototype Validation • Impact

PRELIMINARY SCREENING
PROTOTYPE / PoC

1 SENSING & PROCESSING METHODOLOGY

01 BREATH SAMPLING
Non-invasive breath enters a replaceable mouthpiece and controlled air-flow chamber.

02 MULTI-SENSOR FUSION
BME688 AI Gas Sensor + MQ-series sensors capture VOC and gas-response fingerprints.

03 EDGE ACQUISITION
ESP32 samples sensor signals and performs local filtering, baseline correction and normalization.

04 AI FEATURE ANALYSIS
BSEC2-assisted gas features feed an embedded pattern-classification pipeline at the edge.

05 RISK OUTPUT
The interface presents a rapid screening indication with confidence and sensor telemetry.

STAKEHOLDER VALIDATION & IMPACT

Prototype reviewed in a dedicated demonstration with West Bengal Police, with emphasis on operational necessity, ease of use and rapid response. Target response: ~15 seconds. Indicative unit cost: under INR 3,000, positioning the device as a scalable, locally engineered preliminary screening tool.

2 SYSTEM BLOCK DIAGRAM

HUMAN BREATH INPUT
Mouthpiece • airflow chamber

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MULTI-SENSOR ARRAY
BME688 + MQ2 + MQ3 + MQ135

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ESP32 EDGE CONTROLLER
ADC / digital acquisition • Embedded C

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SIGNAL PROCESSING
Filtering • baseline • normalization

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AI PATTERN CLASSIFIER
Feature fusion • screening decision

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RESULT & DASHBOARD
Risk indication • confidence • report / wireless

EDGE-FIRST DESIGN

Core processing is local, enabling rapid screening without dependence on continuous internet connectivity.

3 PROOF OF CONCEPT



HANDHELD DEVICE + ARCHITECTURE





LIVE SOFTWARE INTERFACE

PROJECT DEMO QR

KEY FEATURES

Non-invasive • Portable • Multi-sensor • AI-assisted
Rapid screening • Wireless data • Digital logs

IMPORTANT: Intended as a preliminary screening / research prototype. Positive indications require appropriate medical, laboratory or forensic confirmation.